Subject / Theme: Surface chemistry and Colloids

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→ Definition of colloids
→ Dispersion, Dispersed medium, dispersed phase
→ Type of colloids
→ Application of colloids
→ Properties of Sol.
→ Lyphillic sol, lyphobic sol (Defn, difference)
→ Gel and emulsion.

* Colloids:-

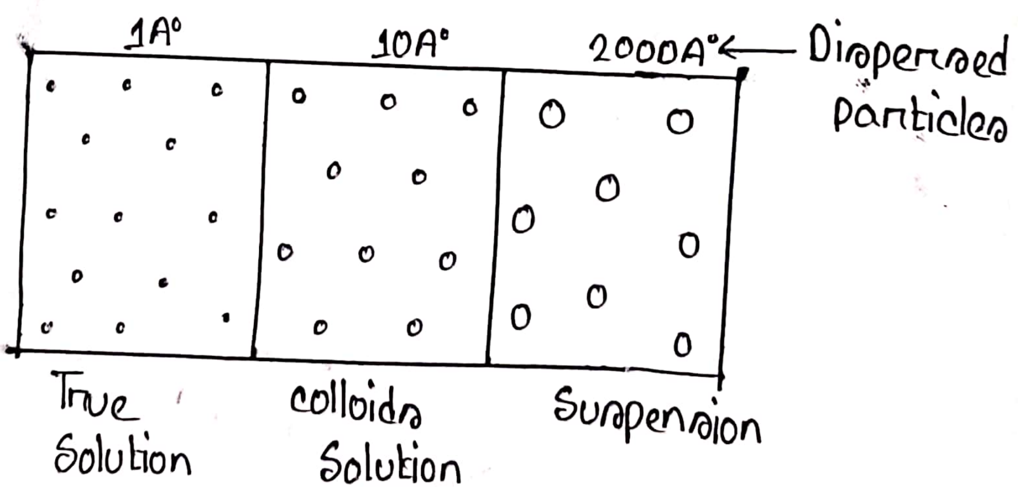
when the diameters of the particles of a substance dispersed in a solvent ranges from about 10A° to 200A° the system is termed as colloids.

Sol:-

Sol is a colloidal suspension with solid particles. is as

True solutions: Salt/Sugar in water (1-10) A°

Suspension: Sand Stirred into water



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A colloidal system made of two phases:-

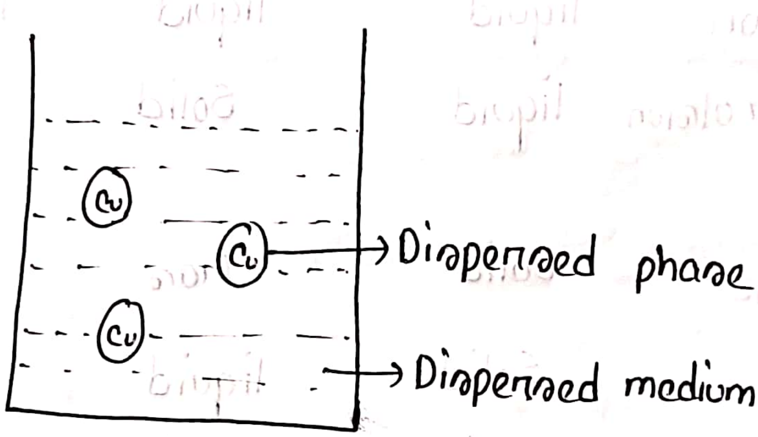
1. Dispersed Phase
2. Dispersed Medium.

*** Dispersed Phase:-**

The substance distributed as colloidal particle is called dispersed phase.

*** Dispersed Medium:-**

The second continous phase in which the colloidal particles are dispersed, is called the dispersed medium.



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* Types of colloids :-

classification of colids based on dispersed medium and dispersed phase:-

Type Name	Dispersed Phase	Dispersed medium	Example
Foam	Gas	liquid	Soda water, whipped cream, shaving cream
Solid Foam	Gas	Solid	Foam Rubber.
Aerosol	liquid	Gas	Mists, clouds.
Emulsion	liquid	liquid	Milk, hair cream
Solid emulsion (gel)	liquid	Solid	butter, cheese
Smoke	Solid	Gas	dust, soot in the air
Sol	Solid	liquid	Paint, ink, colloidal gold.
Solid Sol	Solid	Solid	Ruby. glass, alloys.

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Application of ~~collids~~:- colloids:-

1. A colloids is used as thickening agents in industrial products such as lubricants, lotions, toothpaste etc.

2. In the manifacture of paints and inks, colloids are useful.

3. In ball point pens, the ink is a gel (liquid-solid colloids).

4. Most of the medicines are colloidal

→ Colloidal Gold and Calcium (vitality of muscles)

→ Argyral is used as an eyelotion.

→ Albumin, Hetastrach, Dextran.

5. Colloids are used in water purification.

6. Rubber is obtained by a colloidal Solution.

7. Micelles formed in the cleaning action of soaps are colloids.

8. Many nano materials are prepared by colloids.

9. Silver colloids is used as a germicidal agent.

10. It is used in plastic surgery of human body.

11. These are used in wound dressing materials.

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* Lyophilic Sols (Solvent loving):-

Lyophilic sols are those in which the dispersed phase exhibits a define affinity for the medium or solvent.

Example:- Straach/protein – in H_2O.

* Lyphobic Sols (Solvent hating):-

lyophobic sols are those in which the dispersed phase has no attraction for the medium or the solvent.

Example:- Gold/Sulpher, $Fe(OH)_3$ in H_2O.

* Lyophilic Sols vs Lyophobic Sols:-

Lyophilic Sols	Lyophobic Sols.
1. Prepared by direct mixing with dispersion medium.	1. Cannot be prepared by direct mixing with dispersion medium, its required method.
2. Not little charges in particles.	2. Particles carry positive or negative charge.
3. It doesn't exhibit tyndall effect	3. It exhibits tyndall effect.
4. Reverisible	4. Irreversible
5. More hydrated.	5. less hydrated.
6. More stable	6. Less stable.

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7. Example: Strach or protein dissolved in water.	7. Example: Ferric hydroxide/ Aluminum hydroxide dissolve in water.

* Emulsion:-

It is a liquid-liquid colloidal system. It can be defined as dispersion of finely divided liquid droplets in another Liquids.

Example: Milk, Lotion.

Two types of emulsion:

1. Water in oil type.
2. Oil in water type.

* Role of Emulsion:-

1. Emulsion are used in topical products like lotions and creams to help drugs and cosmetic agents to penetrate the skin.
2. Emulsions are used to blend ingredients that would otherwise seperate, contributing to the texture, appearance, and sensory experience of food products. Mayonnaise is an example of an oil in water emulsion.
3. Emulsions are also used in firefighting.
4. Surfactants in laundry agents and household cleaners emulsify oily dirt particles so they can be washed away.

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5. Bile in the small intestine acts as an emulsifier to help lipase to break down dietary fats.

6. Emulsion paints is a water-based paint that can be used on walls and ceilings.

* Absorption vs chemisorption:-

Absorption	Chemisorption
1. The forces operating in these are weak vander waal's force.	1. The force operating in these cases are similar to those of a chemical bond.
2. The heat of absorption is low i.e about 20-40 kJ mol^{-1}.	2. The heat of chemisorption is high i.e 40-400 KJ mol^{-1}.
3. No compound formation take place.	3. Surface compound are forme
4. The process is reversible	4. The process is irreversible
5. It doesn't need any activation energy.	5. It requires activation energy.
6. It forms multi molecular layer.	6. It form uni-molecular layer.

* Absorption Isotherm:-

The process of absorption is usually studied through graphs which are known as absorption Isotherm.

* Langmuir's Absorbtion Isotherm:-

The rate of absorption (R_a) is proportional to the available surface ($1-\theta$) created by the absorbate and the pressure.

$$R_a \propto (1-\theta)P$$

$$\Rightarrow R_a = K_1(1-\theta)P \quad [K_1 = \text{constant}]$$

Now rate of disorption $\propto \theta$.

$$= K_2\theta \quad [K_2 = \text{constant}]$$

$$\therefore K_1(1-\theta) = K_2\theta$$

$$\Rightarrow K_1P - K_1\theta P = K_2\theta$$

$$\Rightarrow K_2\theta + K_1\theta P = K_1P$$

$$\Rightarrow \theta = \frac{K_1P}{(K_1P + K_2)}$$

Now, $$\frac{P}{W} = \frac{1}{A} + \frac{B}{A}P$$

Where, $$A = \frac{K_1K_2}{K_2} = \text{constant.}$$

$$B = \frac{K_1}{K_2} = \text{constant.}$$

Freundlich Absorption Isotherm:-

Freundlich proposed an emphirical relation in ~~which~~ the form of a mathematical equation-

$$\boxed{\frac{W}{m} = Kp^{1/n}}$$

Where,

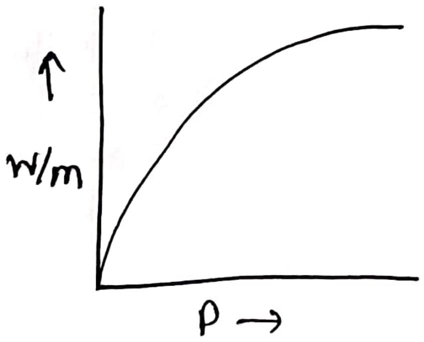
W = mass of ~~gas~~ the gas absorbed.

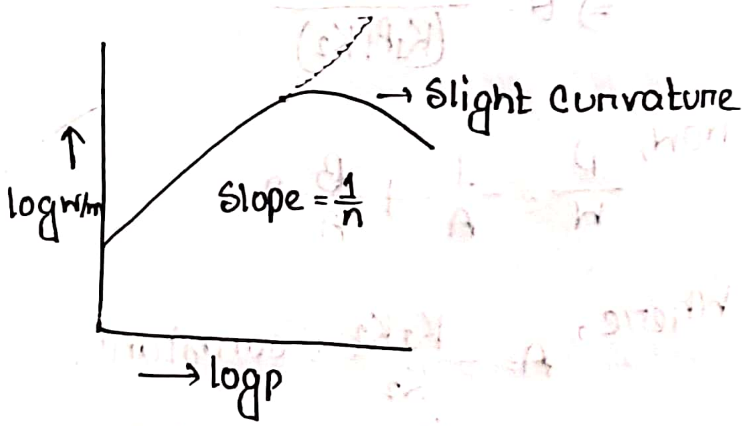
m = mass of absorbent.

P = pressure.

K & n = constant.

This relation is generally represented in the form of a curve obtained by plotting the mass of the mass of the gas ~~t~~ absorbed per unit ~~g~~ mass of absorbent (W/m) against equilibrium pressure.





$$\log\frac{W}{m} = \log K + \log p^{1/n}$$

$$\Rightarrow \log\frac{W}{m} = \log K + \frac{1}{n}\log P.$$

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This equation for a straight line. Thus a plot of $\log \frac{w}{m}$ against $\log P$ should be a straight line with slope $\frac{1}{n}$ and intercept $\log K$.

However, it is found that the plots were straight lines at a low temperature. This indicates that Frendlich equation is approximate and does not apply to ~~af~~ absorption of gases by solids at higher pressure.